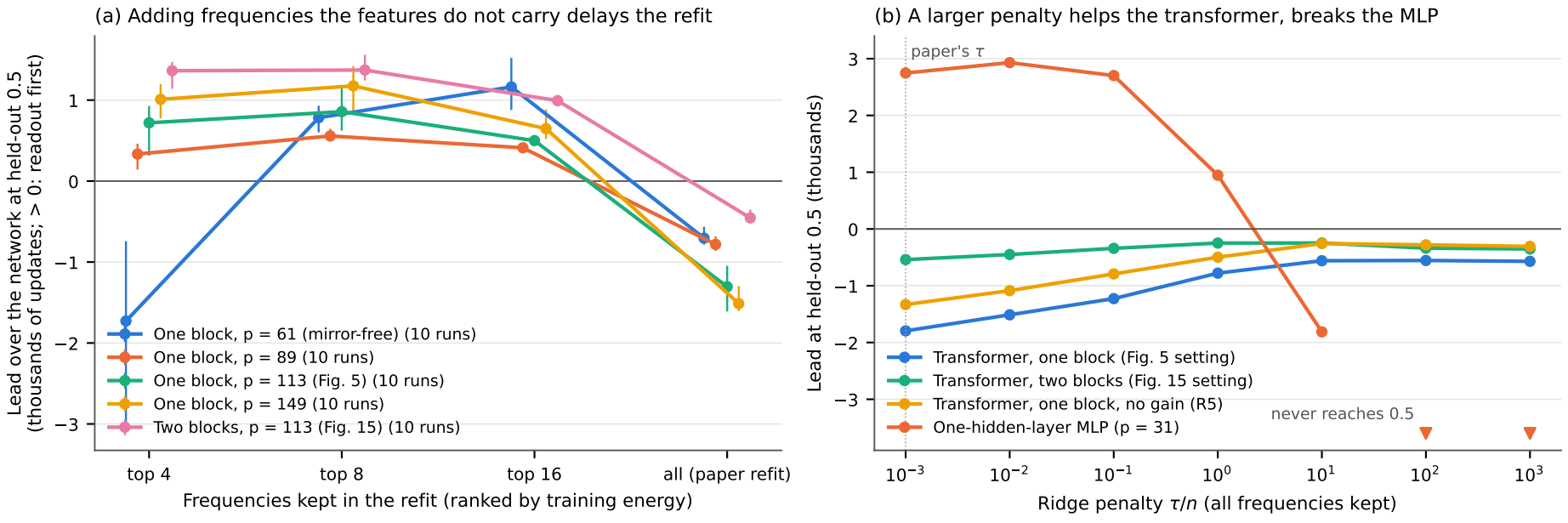
(a) Adding frequencies the features do not carry delays the refit
Lead over the network at held-out 0.5
(thousands of updates; > 0: readout first)
1
0
−1
−2
−3
One block, p = 61 (mirror-free) (10 runs)
One block, p = 89 (10 runs)
One block, p = 113 (Fig. 5) (10 runs)
One block, p = 149 (10 runs)
Two blocks, p = 113 (Fig. 15) (10 runs)
top 4
top 8
top 16
all (paper refit)
Frequencies kept in the refit (ranked by training energy)
(b) A larger penalty helps the transformer, breaks the MLP
Lead at held-out 0.5 (thousands)
3
2
1
0
−1
−2
−3
paper's τ
Transformer, one block (Fig. 5 setting)
Transformer, two blocks (Fig. 15 setting)
Transformer, one block, no gain (R5)
One-hidden-layer MLP (p = 31)
never reaches 0.5
10^−3
10^−2
10^−1
10^0
10^1
10^2
10^3
Ridge penalty τ/n (all frequencies kept)